Félix Wilhelmy

Software Engineer

Education

Master's Degree with Thesis 2024–2026
École de Technologie Supérieure
Information Technology Engineering

Graduate Coursework 2025
MILA / Université de Montréal
Representation Learning (IFT6135)

AlgoLab Summer School 2024
Institut Quantique / Université de Sherbrooke
Near-Term Quantum Algorithms

Bachelor of Engineering 2018–2024
École de Technologie Supérieure
Software Engineering

DEC 2013–2016
Collège Montmorency
Computer Science

Skills

Programming Languages

Python, C++, C#, Java, SQL, TypeScript, JavaScript

Tools

Angular, React, ASP.NET, Entity Framework, NumPy, Pandas, Git, Docker, Linux, Bash, SSH, Jenkins, Pytest

Software Engineering

Data Structures, Algorithms, DevOps, CI/CD, Unit Testing, Software Architecture, Performance Optimization

Application & Systems Development

Full-Stack Development, Backend Development, Database Optimization, Asynchronous Programming, Networking

Data Engineering & ML

Data Pipelines, Data Modeling, Data Quality, Model Training, Model Evaluation, Statistical Analysis

Languages

French (Native)

English (Professional Proficiency)

Contact

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About

I just completed my research master's in information technology engineering (GPA 4.06/4.30; Excellent mention at my thesis defense). I combine industrial software engineering with hands-on academic research, taking complex systems from initial concepts to working solutions. My experience includes technical leadership and international technical competitions. A scientific paper based on my master's research is in preparation.

Experience

QB-PIER – Automated Model Security Testing Framework 2024–2026
Master's Research Project - École de Technologie Supérieure

- Architected and built a modular testing framework from scratch to automate adversarial security evaluations, model training, and comparative analysis.
- Designed configuration-driven workflows, experiment tracking, checkpoint management, and reproducible reporting, organizing over 30 GB of artifacts across 50 trained model variants.

Esports Data Analyst May 2026–Sep 2026
Canadian National Team - Overwatch World Cup 2026

- Supported Team Canada's coaching staff with performance analysis, opponent scouting, and data-driven recommendations for international competition
- Built Scrimlytics with Python, SQL, and DuckDB, transforming high-volume match logs into structured metrics, interactive dashboards, and composition comparisons
- Implemented reusable analytics services, data-quality checks, automated tests, and recurring HTML/PDF reports to support coaching decisions

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Full Stack Developer (Internship) — React/C# Sep 2022–May 2023
Amilia - Montréal, QC

- Developed and maintained customer-facing features in a production SaaS application, including gift-card functionality across C# backend services and a React frontend
- Improved software reliability and developer workflows through unit testing, Jenkins CI/CD enhancements, debugging, and onboarding documentation

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Research Assistant (Internship) — C++/Networking Apr 2021–Aug 2021
LASSENA Research Laboratory - Montréal, QC

- Developed cross-platform C++/Qt research software for avionics analysis, including aircraft instrument interfaces and navigation-data integration
- Designed UDP-based communication for instrument data exchange and optimized background processing to maintain responsive interfaces under real-time constraints

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Full Stack Developer (Internship) — Angular/C# May 2019–Sep 2019
Air Transat - Montréal, QC

- Accelerated a critical SQL search query by approximately 6× through Entity Framework optimization, lighter query models, and asynchronous processing
- Developed internal employee flight-request and availability tools in C#/Angular; migrated Angular 7.2 to 8 and introduced lazy-loaded modules

Awards

1st Place, iQuCodeFest 2025
Polytechnique Montréal
Computing Hackathon

2nd Place, YQuantum 2025
Yale Quantum Institute
Optimization Hackathon

Best Effort Award, Mil'HaQ Fest 2025
MILA / Université de Montréal
Machine Learning Hackathon

Projects

Adaptive Graph Strategy (Hackathon) 2026
MIT iQuHack 2026 - Cambridge, MA, USA

- Co-developed a modular decision and execution system for a graph-based quantum-network challenge with probabilistic outcomes and limited resources
- Integrated reusable circuit components, automated edge selection, outcome-driven feedback, and execution logging into a common Python workflow

Wildfire Resource Allocation (Hackathon) 2025
Yale YQuantum 2025 - New Haven, CT, USA

- Co-developed a resource-allocation tool for wildfire response using geospatial risk data and constrained optimization
- Built a configurable Python pipeline for GeoTIFF processing, geographic cropping, spatial downsampling, optimization, and result visualization

FR2SQL – French Text-to-SQL 2025
Graduate Coursework - École de technologie supérieure

- Developed a French-language Text-to-SQL prototype connecting natural-language queries with relational database schemas through semantic matching and LLM prompting
- Built Python/SQLite schema inspection, Sentence Transformers schema linking, and Hugging Face inference and Spider-FR evaluation tooling

Additional Experience

President (24–25) & VP Competitions (23–24) Sep 2023–Jan 2026
Student Club QuantumÉTS - Montréal, QC

- Led a 40-member quantum computing club, secured its first industry sponsorship, and coordinated technical workshops, competition teams, and student mentorship
- Guided teams achieving podium finishes at all six competitions attended during my presidency, including hackathons hosted by MIT, Yale, MILA, and Polytechnique Montréal

Teaching Assistant Jan 2024–Aug 2025
École de Technologie Supérieure - Montréal, QC

- Guided quantum-computing labs, helped students implement and debug algorithms, and delivered an introductory machine-learning guest lecture

Research & Development (Internship) — Machine Learning 2016
National Film Board of Canada - Montréal, QC

- Developed Python/Tesseract OCR tools supporting model training, evaluation, and research experimentation